

An honest assessment: active inference as the basis for CZB quantification

2026-08-27, written at Jonas's request, ahead of discussing the fitting-plan questions. This is an opinion document; where it states judgments they are mine and hedged as such. Numbers are from this repository's committed analyses (replication/czb/pilot_surface_fit.py and the comparison run recorded in section 3; replication/causation/; notes/05_validation.md).

0 The verdict in one paragraph

As I read the evidence, the active-inference route is a **well-posed, falsifiable, and currently promising research program** for quantifying comfort-zone boundaries — the first human-data contact (a two-parameter threshold on the zero-refit field reproducing the 18-cell cut-in response surface with correlation 0.90) is genuinely encouraging — but its distinctive value is **not yet demonstrated**, because that value lives entirely in the cross-scenario transfer claim, which is untested. Within a single scenario the field is beaten by a trivial scenario-specific regression (section 3), so the method must stand or fall on what the regression cannot do: carry one interpretable number per driver across scenarios. For the ADAS-triggering purpose specifically, my feeling is that the framework is the right *shape* for the problem (one dial, percentile semantics, decomposable triggers, cheap online evaluation) while being some distance from a deployable recipe: the fitted level absorbs paradigm and latency offsets that matter for absolute trigger timing, the field's location depends on stated conventions, and the population behind any percentile is presently 43 crowdsourced participants. The honest unique selling point is not "active inference" as a brand — most of that machinery is switched off in our use — but the **dual role of one quantity**: the same scalar whose accumulation explains *when* drivers act is the scalar whose level set *defines* the boundary. No conventional indicator has that property, and it is testable.

1 What is actually being used, against what the label suggests

"Active inference" names a large theoretical stack. The CZB method deliberately uses a thin slice of it, and it seems to me important to be plain about which slice, because it determines what a skeptical reviewer can and cannot attack.

component of the full framework	used for CZB?	role in our method
preference distribution $p(o)$	yes — central	the comfort-zone field is its pointwise residual information; the boundary is a level set
residual information (surprise family)	yes	turns log-preference into a non-negative scalar with a true zero ("comfortable" = 0)
evidence accumulation on pragmatic value	yes (stage 2)	links the field to response <i>timing</i> ; Markkula-style, not uniquely active-inference
expected free energy planning, policy search	no	the field method is deliberately rollout-free; that is what makes it cheap
epistemic value / information seeking	no	$\alpha = 0$ in the validated configuration; the gaze half is unexercised
belief updating, state estimation	no	the field is evaluated on observed kinematics directly

Two consequences. First, a fair description of our method is "a satisficing preference field with an accumulator, inherited from a validated active-inference driver model" — the theory supplies the *specific calibrated form* and the connection to a model that also produces closed-loop behavior, not the day-to-day machinery. Second, the criticism "you could do this with a hand-designed cost function" is, as I read it, technically true and practically empty: we would then have to invent the cost function, and the whole point is that this one arrives pre-shaped and pre-calibrated by someone else's behavioral validation, with its parameters documented and its response-timing predictions testable on the same data. The convergence with the QUADRARUM document's definition of the CZB — the point where the satisficing condition is violated — is real: a preference distribution *is* a specification of acceptable outcomes.

2 The evidence ledger, as it stands today

claim	evidence	my reading of its strength
the field recovers response onsets from kinematics alone	median onset error 0.0 s against the authors' OSF deposit (n = 896, score 0.855)	strong, but model-recovering-model — no human in the loop

claim	evidence	my reading of its strength
the field explains a human response surface	stage-0 pilot: 2-parameter probit, corr 0.90 / RMSE 0.125 on the 18-cell cut-in surface	encouraging first human contact; see section 3 for the honest context
the response process is behaviorally right	crash-causation study: active-inference conditions closer to the QUADRIIS reference than the CBM control (severity θ 0.148 vs 0.209, $P \approx 0.97$; braking aggregate 0.3% vs 35%)	moderate; supports the <i>response model</i> , and severity constrains timing only weakly (severity-vs-timing dissociation)
one scalar transfers across scenarios	—	untested; this is the load question
the field expresses graded caution in mild events	mild truck cut-ins: dread-field deficit exactly zero through the response window	negative as it stands; the comfort level (allowed deceleration below capability) is required to fix it
the paradigms measure one latent process	fixed-clip CDF vs button density agree within 0.06 at high criticality; +0.26 excess early pressing at intermediate	mixed; consistent with one process plus a shift, but the shift is real and confounded with order

3 The uncomfortable comparison, run rather than assumed

If the field is only a repackaging of kinematics, a kinematic baseline should match it. I fitted probit models to the same 18-cell surface (3 096 trials) at matched or nearly matched parameter counts:

model	parameters	RMSE	corr
comfort-zone field (running max), threshold + noise	2	0.125	0.903
TTC at clip end	2	0.185	0.764
1/TTC at clip end	2	0.175	0.792
gated required deceleration (running max)	2	0.203	0.727
nominal TTC + exposure time (design variables)	3	0.081	0.960
1/TTC at end + exposure time	3	0.073	0.968

(Addendum, same evening: Jonas asked whether this comparison is in-sample. The table above is; the check has now been run held-out, and the conclusion survives. Under leave-one-criticality-out cross-validation — fit on two criticality levels, predict the third — the field gives out-of-sample corr 0.864 / RMSE 0.147 against the design regressions' 0.934–0.949 / 0.097–0.108, and leave-one-timepoint-out gives the same picture. The regressions genuinely generalize better within this scenario; overfitting is not what made them win. Committed in replication/czb/pilot_surface_fit.py.)

Three readings, in decreasing comfort:

- The field beats every single kinematic scalar at equal parameter count.** Plain end-TTC cannot explain the surface (it thinks a C1 clip at TTC 4.0 is more dangerous than a C6 clip at TTC 4.5, and drivers say the opposite), because the response depends on the developing lateral geometry — which is exactly what the field's lane-entry structure integrates. To rescue TTC one must gate it by lane relevance, and by then one has rebuilt the field's safety term by hand.
- A scenario-specific regression on the design variables beats the field.** One extra parameter and an explicit time term give 0.96. This is not embarrassing — a regression fit to *this* scenario's design axes cannot transfer, predicts nothing outside its grid, and its coefficients mean nothing — but it disciplines the claim: the field cannot be sold on within-scenario fit. Its case is that its *threshold* is one portable, interpretable number, where the regression's coefficients are neither.
- The gap is the accumulator's signature.** Most of the baseline's advantage comes from the exposure-time term (TTC alone 0.76 → 0.96 with time added). The static threshold cannot express "probability of having responded grows with time at a given criticality"; the accumulator exists to express exactly that, without adding per-scenario parameters. My expectation — falsifiable, stage 2 — is that field + accumulator closes most of this gap; if it does not, that is evidence against the framework worth taking seriously.

4 What the route genuinely buys, if the transfer test passes

- A one-dimensional population.** The percentile question ("trigger where 80% of drivers would already be uncomfortable") becomes well-posed without an ellipse, because the field has already absorbed speed, gap,

closing rate and lateral position into one number. The multivariate conjunction problem the QUADRARUM document works through is dissolved rather than solved.

- **The dual role.** The same scalar defines the boundary (level set) and predicts response timing (accumulation). A TTC threshold does not predict *when* in a developing event a driver acts; this does, and the button data tests it.
- **Decomposable triggers.** Exceedance attributes to named terms (speed, clearance, counterfactual braking), which matters for explaining an ADAS intervention to an engineer, a regulator, or a driver.
- **Cheap online evaluation.** The field is closed-form in observables; no rollouts. It could run in a vehicle at trivial cost, unlike the full closed-loop model (~18 s CPU per simulated step).
- **A benefit-estimation synergy.** The crash-causation machinery in this repository can, in principle, estimate the *safety* consequence of any candidate trigger placed on the field — the two workstreams compose into "set the trigger by comfort percentile, audit it by counterfactual crash outcome", which neither piece does alone. To me this composition is one of the strongest practical arguments for keeping both on the same scalar.

5 The case against, stated as risks with severities

- 1 **The transfer claim may simply be false** (severity: decisive; probability: real). Drivers may not carry one level across scenario types — the instruction differs by scenario in our own data (intervene / abort / yield), and handbook ch. 04 already flags that pooling these under one scalar is the likeliest way to manufacture a false negative. The fallback (elliptical joint percentile on conventional observables) is tracked, which is the right structure: failure would be a finding, not a dead end.
- 2 **What transfers is the threshold, not the work** (severity: moderate, chronic). Every new scenario needs its field construction — geometry, norms, observation conventions. The very first new scenario (cut-in) required a structural modification to the preference function, argued but still a modification made before any fitting; LTAP will need a crossing geometry; the overtake traces need role assignment rebuilt. "One scalar, no per-scenario engineering" is not the true position; the true position is "bounded, principled, parameter-free per-scenario construction with a shared threshold", and it should be stated that way.
- 3 **The fitted level is not the boundary** (severity: moderate for percentiles, high for absolute triggers). $c_{\text{obs}} = c_{\text{true}} + \text{bias} + \text{paradigm} + \text{latency}$. The design identifies some components (C1 for bias, framing contrast) and not others (motor latency, unidentified; the paradigm term is confounded with order). Percentiles largely survive because the offsets are shared; an absolute trigger time does not — 0.2 s of unmodeled latency is 6 m at highway speed.
- 4 **The field's location rests on stated conventions** (severity: moderate, manageable). The assumed worst case $a_{\text{OV,min}}$, the reaction budget t_{react} , and the capability a_{max} move the boundary substantially — at 20 m/s the comfort-to-dread span (allowed deceleration 4 vs 8 m/s²) is a time-headway range of 1.86 s down to 0.61 s, a factor of three. The project's standing rule (never quote absolute boundaries without stating them) applies to any deployed threshold verbatim. Freezing them across scenarios is part of the claim under test; tuning them per scenario would readmit the multivariate problem through the back door.
- 5 **Hard zeros below the counterfactual threshold** (severity: high if unaddressed; fix identified). The dread field is exactly silent for the mild truck cut-ins while humans plausibly still respond. An ADAS triggered on that field would never fire on mild-but-uncomfortable events. The comfort-level formulation (allowed deceleration below capability) exists precisely for this; the fitting plan's section 4 makes the truck cells the internal check. If drivers' responses there turn out to need a different *shape*, not just a lower level, that would be structural evidence against the field.
- 6 **Provenance** (severity: low-moderate, documented). The preference function follows the authors' released code, which deviates from its own supplementary information in 13 documented ways, and the paper's worked example does not match its deposit. We build on the implementation that generated the validated results — the right choice, I think — but a safety-adjacent application inherits that provenance and should say so.
- 7 **Population and paradigm** (severity: high for deployment, low for the research question). Any percentile from this data describes 43 crowdsourced participants watching clips, with a demonstrated satisficing excess (+0.26 at intermediate criticality) in one paradigm. This supports method development, comparisons and contrasts; it does not support setting a fleet trigger.

6 Specifically as a way to place an ADAS trigger

The end-to-end pipeline would be: fit c per driver → population distribution of c → choose a percentile → the trigger surface is that level set of the field, evaluated online. Where I would concentrate skepticism, in order:

- 1 **The percentile choice dominates.** Moving the percentile moves the trigger far more than most estimation errors will; the sensitivity of the implied trigger timing to the percentile (and to the comfort-versus-dread level choice) should be quantified *before* investing further in estimator refinement, because it may show that a cruder estimate suffices — or that no defensible percentile gives a usable trigger. This is cheap to run once stage 1 exists.
- 2 **Comfort is an acceptance criterion, not a safety criterion.** A CZB-percentile trigger answers "when would most drivers already want regulation", which is the right question for nuisance and trust, and only indirectly a safety question. The QUADRARUM document's observation that measured thresholds and CZB-based ones may differ by only ~ 0.1 s suggests the two may nearly coincide in practice, but that is a hypothesis to check — and the causation machinery here can check it, by scoring candidate triggers against counterfactual crash outcomes on the QUADRIS ensemble.
- 3 **Absolute timing inherits every offset in risk 3.** For a comparative use (rank situations, adapt sensitivity between drivers) the route is already fairly solid; for an absolute "fire now" threshold the latency and paradigm terms must be fixed externally, and the honest statement is that this design cannot identify them.

7 Bottom line, and what would change my mind

My overall position: **worth pursuing, with the transfer test as the gate, and worth describing more modestly than "active inference for CZB"**. The within-scenario evidence says the field is a good — not the best — single predictor, distinguished by interpretability, portability claims, and its link to response timing rather than by raw fit. The framework earns its keep if and only if (a) the accumulator closes most of the within-scenario gap without per-scenario parameters, and (b) a level fitted on cut-in predicts the other three scenarios better than chance and comparably to per-scenario refits. If both hold, this is, to my knowledge, the first one-dimensional, testable, driver-population CZB — a real contribution. If (b) fails, the elliptical formulation on conventional indicators is the honest fallback and the field reverts to being one indicator among several. Concretely decidable, on data we already hold, within the current fitting plan — which is, I think, the strongest thing that can be said for it.